Supplemental Material for "Winding-free exact diagonalization of quantum spin ice: benchmarks and convergence criteria"

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MODEL, CONVENTION, AND CLUSTER GEOMETRY

We use local spin axes and

$$H = H_0 + V = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-). \tag{1}$$

Huang *et al.* use this same convention at the sign-free point $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$ [1]. No sign or factor-of-two conversion is therefore needed for the external benchmark. For the diagonal local-axis form $J_x S_i^x S_j^x + J_y S_i^y S_j^y + J_{zz} S_i^z S_j^z$, our convention is

$$J_\pm = -\frac{J_x + J_y}{4}, \qquad J_{\pm\pm} = \frac{J_x - J_y}{4}. \tag{2}$$

Sites are represented in a periodic covering lattice. For each nearest neighbor pair (i, j), the stored integer image $\boldsymbol{n}_{ij}$ gives the lifted bond displacement

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}^T L, \tag{3}$$

where the rows of L are the three supercell vectors. A depth-first cycle enumeration retains simple cycles that preserve the ice constraint and, for six-cycles, are planar in the covering lattice. The resulting census is listed in Table I. There are no contractible four-cycles. The active geometry module performs the enumeration independently

TABLE I. Finite-cluster ice-space and loop census.

cluster	rank $P_{\rm ice}$	winding 4	contractible 6	wrapping 6
cubic-16	90	36	16	48
FCC-32	2970	24	32	96

from the archived campaigns.

ICE-RULE PROJECTOR AND MICROSCOPIC SOURCE

Define the tetrahedron charge $\widehat{Q}_t = \sum_{i \in t} S_i^z$. Since the charges commute, they have a common S^z basis. The symbol $\mathbf{1}_{\{0\}}(\widehat{Q}_t)$ means the projector onto the zero-charge eigenspace of tetrahedron t:

$$\mathbf{1}_{\{0\}}(\widehat{Q}_t)|s\rangle = \begin{cases} |s\rangle, & Q_t(s) = 0, \\ 0, & Q_t(s) \neq 0. \end{cases} \tag{4}$$

For spin 1/2, $Q_t \in \{-2, -1, 0, 1, 2\}$, so the same local projector has the explicit polynomial form

$$\mathbf{1}_{\{0\}}(\widehat{Q}_t) = (1 - \widehat{Q}_t^2)(1 - \widehat{Q}_t^2/4). \tag{5}$$

Multiplying these commuting local projectors imposes zero charge on every tetrahedron:

$$P_{\rm ice} = \prod_t \mathbf{1}_{\{0\}}(\widehat{Q}_t) = \sum_{\substack{s:\, Q_t(s)=0 \\ \text{for every } t}} |s\rangle\langle s|. \tag{6}$$

This definition is independent of cluster size and of any enumeration of ice configurations. The operator is Hermitian and idempotent, $P_{\rm ice}^\dagger = P_{\rm ice}$ and $P_{\rm ice}^2 = P_{\rm ice}$, and projects onto the full subspace satisfying the ice rule on every tetrahedron.

For an oriented bond (i, j), the stored image $\boldsymbol{n}_{ij}$ defines

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i - \boldsymbol{n}_{ij}^T L, \qquad \boldsymbol{c}_{ij} = \boldsymbol{r}_i + \boldsymbol{r}_j - \boldsymbol{n}_{ij}^T L. \tag{7}$$

This second coordinate is fixed, rather than chosen as an additional source convention. Write the lifted position of the second site as $\widetilde{\boldsymbol{r}}_j = \boldsymbol{r}_j - \boldsymbol{n}_{ij}^T L$. Since S^+ changes S^z by $+1$, a double-raising move changes the lifted spin polarization by

$$\Delta\boldsymbol{p}(S_i^+ S_j^+) = \boldsymbol{r}_i + \widetilde{\boldsymbol{r}}_j = \boldsymbol{c}_{ij}, \tag{8}$$

whereas double lowering gives $-\boldsymbol{c}_{ij}$. Similarly, $S_i^+ S_j^-$ gives $\boldsymbol{r}_i - \widetilde{\boldsymbol{r}}_j = -\boldsymbol{d}_{ij}$. Attaching $e^{-2i\boldsymbol{\theta}\cdot\Delta\boldsymbol{p}}$ to each matrix element therefore produces all four phases below from one polarization-counting rule. The source is inserted directly into the microscopic exchange terms:

$$\begin{aligned} H(\boldsymbol{\theta}) = H_0 &- J_\pm \sum_{\langle ij\rangle} \left[e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^+ S_j^- + e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}} S_i^- S_j^+ \right] \\ &+ J_{\pm\pm} \sum_{\langle ij\rangle} \left[e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^+ S_j^+ + e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}} S_i^- S_j^- \right]. \end{aligned} \tag{9}$$

Equation (9) is an exact microscopic finite-cluster Hamiltonian. No low-energy reduction, energy denominator, or expansion in $J_\pm/J_{zz}$ or $J_{\pm\pm}/J_{zz}$ has been used. For a basis state with i down and j up, the corresponding off-diagonal matrix element is

$$\langle S_i^+ S_j^- s | H(\boldsymbol{\theta}) | s\rangle = -J_\pm e^{+2i\boldsymbol{\theta}\cdot\boldsymbol{d}_{ij}}, \tag{10}$$

and the reverse exchange carries the complex-conjugate phase. This is the complete mathematical meaning of "attaching a microscopic source." For example, if both spins are down,

$$\langle S_i^+ S_j^+ s | H(\boldsymbol{\theta}) | s\rangle = J_{\pm\pm} e^{-2i\boldsymbol{\theta}\cdot\boldsymbol{c}_{ij}}, \tag{11}$$

and the double-lowering move again carries the complex conjugate. Unlike the $J_\pm$ exchange, the $J_{\pm\pm}$ pair flip changes total S^z by two. The cubic-16 implementation therefore uses the complete 2^{16}-state basis when $J_{\pm\pm} \neq 0$; the 12,870-state $S^z = 0$ block is used only on the $J_{\pm\pm} = 0$ benchmark slice. The code raises an error if a nonzero pair-flip term is requested in a basis that is not closed under it.

For a sequence γ of microscopic moves, step ℓ acts on the stored oriented bond (i_ℓ, j_ℓ). Define its lifted polarization increment

$$\Delta\boldsymbol{p}_\ell = \begin{cases} -\boldsymbol{d}_{ij}, & T_\ell = S_i^+ S_j^-, \\ +\boldsymbol{d}_{ij}, & T_\ell = S_i^- S_j^+, \\ +\boldsymbol{c}_{ij}, & T_\ell = S_i^+ S_j^+, \\ -\boldsymbol{c}_{ij}, & T_\ell = S_i^- S_j^-. \end{cases} \tag{12}$$

The product of source phases is

$$\prod_{\ell\in\gamma} e^{-2i\boldsymbol{\theta}\cdot\Delta\boldsymbol{p}_\ell} = e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}_\gamma}, \qquad \boldsymbol{q}_\gamma = 2\sum_{\ell\in\gamma} \Delta\boldsymbol{p}_\ell. \tag{13}$$

With the primitive coordinate convention used in the code, every completed ice-to-ice sequence has $\boldsymbol{q}_\gamma \in \mathbb{Z}^3$. Contractible paths have $\boldsymbol{q}_\gamma = 0$; the targeted paths carrying nonzero net transport through a periodic image have $\boldsymbol{q}_\gamma \neq 0$. A change of home-cell representatives redistributes phase between bond increments and basis states but does not change the completed transport or its vanishing. Equation (13) follows only by multiplying the microscopic phases and is exact for every exchange sequence; the path notation does not invoke perturbation theory.

The continuous character average of a completed path is consequently

$$\frac{1}{(2\pi)^3} \int_{[0,2\pi)^3} d^3\theta \, e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}_\gamma} = \delta_{\boldsymbol{q}_\gamma,\boldsymbol{0}}. \tag{14}$$

Thus a finite-cell process that closes with nonzero lifted polarization transport has exactly zero winding-free coefficient. A contractible process, including any legitimate pair-flip correction with zero total transport, is retained. On an M^3 grid the right-hand side is instead one for $\boldsymbol{q}_\gamma = \boldsymbol{0} \pmod{M}$, which is the finite-$M$ alias discussed below.

For a concrete mixed-channel example, use the cubic-16 site convention in the code and apply $S_0^- S_1^+$, $S_8^+ S_{10}^+$, and $S_8^- S_{11}^-$. The three lifted increments are

$$\left(0, \frac{1}{4}, \frac{1}{4}\right), \qquad \left(\frac{5}{4}, 0, \frac{5}{4}\right), \qquad \left(-\frac{5}{4}, -\frac{1}{4}, -1\right). \tag{15}$$

The sequence connects two ice configurations but has $\boldsymbol{q}_\gamma = 2\sum_\ell \Delta\boldsymbol{p}_\ell = (0,0,1)$, so Eq. (14) removes this spurious $J_\pm J_{\pm\pm}^2$ four-spin contribution. In contrast, applying a pair flip and its inverse gives increments $-\boldsymbol{c}_{ij}$ and $+\boldsymbol{c}_{ij}$, hence $\boldsymbol{q}_\gamma = 0$; that diagonal backtracking correction is not removed.

Crucially, the protocol does not average the microscopic $H(\boldsymbol{\theta})$ itself. It first obtains the exact source-dependent band and its canonical pullback $h(\boldsymbol{\theta})$, then averages that operator. Every term in a resolvent expansion inherits the phase product above, while the exact Fourier series of $h(\boldsymbol{\theta})$ collects those terms to all orders. Character orthogonality therefore removes the nonzero-transport Fourier coefficients without deleting the microscopic moves needed to build zero-transport paths.

EXACT BAND PROJECTOR AND CANONICAL PULLBACK

The two projectors in the construction have different roles. P_{ice} is the fixed reference projector in Eq. (6). By contrast, $P_{\boldsymbol{\theta}}$ is the exact spectral projector of $H(\boldsymbol{\theta})$ onto the band continuously connected to the ice-rule subspace as the transverse exchange is turned on. If $\Gamma_{\boldsymbol{\theta}}$ encloses this isolated band and no other spectrum,

$$P_{\boldsymbol{\theta}} = \frac{1}{2\pi i}\oint_{\Gamma_{\boldsymbol{\theta}}} (z - H(\boldsymbol{\theta}))^{-1} dz = \sum_{\alpha\in\mathcal{B}_{\boldsymbol{\theta}}} |\psi_{\alpha\boldsymbol{\theta}}\rangle\langle\psi_{\alpha\boldsymbol{\theta}}|. \tag{16}$$

Its rank equals rank P_{ice}. The protocol is undefined if the band touches its complement during the continuation.

The positive operator $G_{\boldsymbol{\theta}} = P_{\text{ice}} P_{\boldsymbol{\theta}} P_{\text{ice}}$ acts on the ice-rule subspace. If it is nonsingular, the polar partial isometry

$$W_{\boldsymbol{\theta}} = P_{\boldsymbol{\theta}} P_{\text{ice}} G_{\boldsymbol{\theta}}^{-1/2} \tag{17}$$

satisfies $W_{\boldsymbol{\theta}}^\dagger W_{\boldsymbol{\theta}} = P_{\text{ice}}$ and $W_{\boldsymbol{\theta}} W_{\boldsymbol{\theta}}^\dagger = P_{\boldsymbol{\theta}}$. It fixes the otherwise arbitrary eigenvector gauge within the exact band. The operator represented in the common ice-rule reference space is

$$h(\boldsymbol{\theta}) = W_{\boldsymbol{\theta}}^\dagger H(\boldsymbol{\theta}) W_{\boldsymbol{\theta}}. \tag{18}$$

For the numerical implementation, let B contain an orthonormal basis of the ice-rule subspace, and let $\Psi_{\boldsymbol{\theta}}$ and $E_{\boldsymbol{\theta}}$ contain the exact band eigenvectors and eigenvalues. Setting

$$X_{\boldsymbol{\theta}} = B^\dagger \Psi_{\boldsymbol{\theta}}, \qquad U_{\boldsymbol{\theta}} = X_{\boldsymbol{\theta}} (X_{\boldsymbol{\theta}}^\dagger X_{\boldsymbol{\theta}})^{-1/2} \tag{19}$$

gives the matrix

$$h(\boldsymbol{\theta}) = U_{\boldsymbol{\theta}} \,\mathrm{diag}(E_{\boldsymbol{\theta}}) U_{\boldsymbol{\theta}}^\dagger. \tag{20}$$

The minimum eigenvalue of $X_{\boldsymbol{\theta}}^\dagger X_{\boldsymbol{\theta}}$ tests the polar overlap. A singular Gram matrix is a failure of the selected-band identification, not a numerical singularity to regularize away.

CHARACTER PROJECTION AND ITS NONPERTURBATIVE CONTENT

The exact pulled-back operator has the transport Fourier series

$$h(\boldsymbol{\theta}) = \sum_{\boldsymbol{q}\in\mathbb{Z}^3} h_{\boldsymbol{q}} e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}}. \tag{21}$$

Its winding-free, zero-transport component is

$$h_{\boldsymbol{q}=0} = \frac{1}{(2\pi)^3} \int_{[0,2\pi)^3} d^3\theta \, h(\boldsymbol{\theta}). \tag{22}$$

The implemented M^3 character rule is

$$h_M = \frac{1}{M^3} \sum_{\boldsymbol{m} \in \mathbb{Z}_M^3} h(2\pi \boldsymbol{m}/M) = \sum_{\boldsymbol{q} \in M\mathbb{Z}^3} h_{\boldsymbol{q}}. \tag{23}$$

The last equality follows from discrete Fourier orthogonality. A finite grid aliases nonzero transports $M\boldsymbol{k}$, so convergence in M is mandatory. In particular, $M = 2$ is only a parity selector: it retains every Fourier component with $\boldsymbol{q} = 2\boldsymbol{k}$. It cannot by itself establish zero transport.

This construction is nonperturbative for the selected finite-cluster band in a precise sense. At every $\boldsymbol{\theta}$, the code diagonalizes the microscopic $H(\boldsymbol{\theta})$ at the target $(J_\pm, J_{\pm\pm})/J_{zz}$ and evaluates $P_{\boldsymbol{\theta}}$, $G_{\boldsymbol{\theta}}^{-1/2}$, and $h(\boldsymbol{\theta})$ without expanding in the coupling. If one introduces a formal parameter λ, analyticity of the isolated band would give

$$h(\boldsymbol{\theta}; \lambda) = \sum_{n=0}^{\infty} \lambda^n \sum_{\boldsymbol{q}} h_{n,\boldsymbol{q}} e^{-i\boldsymbol{\theta}\cdot\boldsymbol{q}}, \tag{24}$$

and Eq. (22) would return $\sum_n \lambda^n h_{n,0}$. The production calculation evaluates the exact left-hand side at $\lambda = 1$ instead of truncating the series on the right.

The claim has explicit boundaries: exact diagonalization is for a finite Hilbert space and chosen symmetry sector; the character integral is reached only after M convergence; and the isolated-band gap and polar overlap must remain open. None of these conditions implies cluster-size convergence.

The order of operations cannot be changed. Averaging $H(\boldsymbol{\theta})$ before diagonalization removes individual hops before they compose. Averaging $\mathrm{Tr}\, e^{-\beta H(\boldsymbol{\theta})}$ averages a nonlinear observable, not the operator in Eq. (22). Likewise, a counterterm guessed before the character projection would cancel selected amplitudes rather than transport sectors. Equation (32) instead embeds the correction only after the exact character projection is complete.

PERTURBATIVE PATH ENUMERATION AS A TOPOLOGY AUDIT

Perturbation theory is not used to construct the production spectrum. It is used to verify Eq. (13) on the shortest known processes. Let $Q_{\text{ice}} = 1 - P_{\text{ice}}$ and

$$R = -Q_{\text{ice}}(H_0 - E_{\text{ice}})^{-1} Q_{\text{ice}}. \tag{25}$$

Because $P_{\text{ice}} V P_{\text{ice}} = 0$, the diagnostic rows at second and third order are

$$h^{(2)} = P_{\text{ice}} V R V P_{\text{ice}}, \tag{26}$$

$$h^{(3)} = P_{\text{ice}} V R V R V P_{\text{ice}}. \tag{27}$$

The code stores each virtual sequence and merges only rows with the same source ice state s_r, target ice state t_r, and covering-lattice image transport. For every off-diagonal row r, the bit mask $s_r \oplus t_r$ identifies the spins flipped by the completed sequence. This mask is matched against the independently enumerated ice-preserving four- and six-cycles. Summing the stored bond images around that cycle classifies it as contractible or wrapping; Eq. (13) independently gives its transported dipole $\boldsymbol{q}_r$.

For a channel $\mathcal{A}$, define a distinct term by its spin-flip support $\mu_r = s_r \oplus t_r$. The number of such terms retained by an M^3 character grid is

$$N_{\mathcal{A}}^{(M)} = \left|\{\mu_r : \ r \in \mathcal{A}, \ c_r \neq 0, \ \boldsymbol{q}_r = \boldsymbol{0} \pmod{M}\}\right|. \tag{28}$$

Here c_r includes the coherent virtual-path sum at fixed (s_r, t_r, image), and $M = 1$ is the unprojected census. The independently generated loop census and row audit give Table II. Thus every audited winding four-loop and wrapping-hexagon term is absent after projection, whereas every contractible hexagon term remains. These support counts are not matrix ranks: rank is a property of the summed operator in a specified basis and need not equal the number of

TABLE II. Number of distinct loop-support terms before and after the finite character projection. A term is counted once for each distinct spin-flip support μ_r; entries under $M = 2, 3, 4$ are $N_{\mathcal{A}}^{(M)}$ from Eq. (28).

cluster	channel	unprojected	$M = 2$	$M = 3$	$M = 4$
cubic-16	winding four-loop	36	0	0	0
cubic-16	contractible hexagon	16	16	16	16
cubic-16	wrapping hexagon	48	0	0	0
FCC-32	winding four-loop	24	0	0	0
FCC-32	contractible hexagon	32	32	32	32
FCC-32	wrapping hexagon	96	0	0	0

geometric terms. The $M = 2$ grid already makes the same classification for these shortest paths, and $M = 3, 4$ leave it unchanged. This does not make $M = 2$ an all-order projector: components with nonzero $\boldsymbol{q} = 2\boldsymbol{k}$ are aliased with zero by Eq. (23). The audit establishes the source's topological meaning at low order; it is not used to construct the exact-band curves.

For reference, canonical Schrieffer–Wolff theory [2] generates the Taylor coefficients $h_{n,\boldsymbol{q}}$ when the band is analytic. Agreement with the exact-band source dependence at weak coupling is a useful audit, but no finite Schrieffer–Wolff order appears in the production definition.

To quantify character-grid convergence without counting a uniform shift of the selected band as a physical change, define

$$\widetilde{h}_M = h_M - \frac{\operatorname{Tr} h_M}{d}\, I_d, \tag{29}$$

$$\epsilon_{M-1\to M} = \frac{\|\widetilde{h}_{M-1} - \widetilde{h}_M\|_{\mathrm{F}}}{\|\widetilde{h}_M\|_{\mathrm{F}}}, \tag{30}$$

where $d = \operatorname{rank} P_{\mathrm{ice}} = 90$ for cubic-16 and $\|A\|_{\mathrm{F}} = [\operatorname{Tr}(A^\dagger A)]^{1/2}$. "Exact" here means that every h_M is averaged from the exact microscopic eigenspaces and their polar pullbacks; it does not mean that a finite M equals the continuous character integral. Centering removes only the identity component, which changes no level spacing, heat capacity, or spectral transition frequency within the band.

One consequence of the continuous character integral is available exactly at finite M. Because $\boldsymbol{q}_\gamma = 2\Delta\boldsymbol{p}$, a matrix element that survives the continuous average cannot change the polarization coordinate, so h_{wf} is block diagonal in $\boldsymbol{p}$. We therefore impose that block structure on h_M, discarding the inter-sector elements that a finite grid retains through $\boldsymbol{q}_\gamma = \boldsymbol{0} \pmod{M}$. On cubic-16 the discarded coupling is $1.6 \times 10^{-5} J_{zz}$ at $M = 4$. This is small in operator norm but not in its consequences: left in place, it splits the sixfold ground-state manifold by $3.5 \times 10^{-5} J_{zz}$, which introduces a spurious Schottky peak at $T/J_{zz} \simeq 1.5 \times 10^{-5}$, drives the residual entropy from $\ln 6$ down to $\ln 3$ below that temperature, and moves spectral weight to $\omega \simeq 0$ at Γ. The splitting tracks the discarded coupling across M and is not a floating-point effect: states within each degenerate multiplet agree to 4×10^{-16}, eleven orders of magnitude below it. Imposing the block structure is exact at any M and removes all three artifacts.

At $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$, Eq. (30) gives 6.36% from $M = 2$ to 3 and 0.013% from $M = 3$ to 4, as shown in Fig. 1. The latter passes the 5% character-convergence gate. The zero-source minimum ice overlap is 0.762, and every generic $M = 3, 4$ point has a larger overlap. These grids are successive character resolutions: $M = 3$ removes the parity alias left by $M = 2$, while comparison with the independent $M = 4$ grid tests the remaining finite-M aliases.

FULL-TEMPERATURE THERMODYNAMICS

The final method is defined on the complete microscopic Hilbert space, not on the ice reference space. The projectors construct its counterterm; they do not restrict the thermodynamic trace. Set $d = \operatorname{rank} P_{\mathrm{ice}}$ and fix the otherwise undetermined scalar placement of the corrected band by

$$\bar{h}_M = h_M + \frac{\operatorname{Tr}[h(\mathbf{0}) - h_M]}{d} I_d. \tag{31}$$

The full-Hilbert winding-free Hamiltonian is

$$H_{\mathrm{wf}}^{(M)} = H(\mathbf{0}) + W_0[\bar{h}_M - h(\mathbf{0})]W_0^\dagger. \tag{32}$$

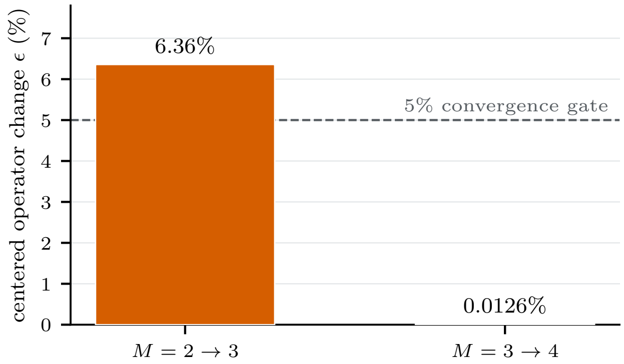

FIG. 1. Successive character-grid convergence of the exact cubic-16 band operator at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$. Bars show the centered relative Frobenius change in Eq. (30); the dashed line is the preassigned 5% gate.

It is one temperature-independent Hermitian operator. Since $W_0 W_0^\dagger = P_0$, it equals $H(\mathbf{0})$ exactly on the spectral complement of the ice-descended band. The correction therefore acts in the microscopic Hilbert space but is generally a nonlocal, band-supported counterterm; the low-order projected rows only verify its loop content.

The partition function is the ordinary full trace

$$Z_{\mathrm{wf}} = \mathrm{Tr}_{\mathcal{H}}\, e^{-\beta H_{\mathrm{wf}}^{(M)}}. \tag{33}$$

For computation, let $\{E_i\}$ be the full periodic spectrum, B the indices of its selected band, and $\{\widetilde{E}_a\}$ the eigenvalues of $\bar{h}_M$. The exact spectral decomposition of Eq. (33) gives, for $k = 0, 1, 2$,

$$M_k^{\mathrm{wf}}(\beta) = \sum_{i \notin B} E_i^k e^{-\beta E_i} + \sum_a \widetilde{E}_a^k e^{-\beta \widetilde{E}_a}. \tag{34}$$

Then

$$E = M_1/M_0, \tag{35}$$

$$C = \beta^2 \left(M_2/M_0 - (M_1/M_0)^2 \right), \tag{36}$$

$$S = \ln M_0 + \beta E. \tag{37}$$

Equation (34) is thus an evaluation identity for the full Hamiltonian in Eq. (32), not a separate thermodynamic prescription. For the present $J_{\pm\pm} = 0$ cubic-16 benchmark we use the complete 65,536-level spectrum. For FCC-32 the same identity requires stochastic trace estimation of the complement; that calculation has not yet been performed.

LONGITUDINAL WINDING-FREE DYNAMICS

The full response is defined using $H_{\mathrm{wf}}^{(M)}$ and the unchanged microscopic spin operator. The plotted range lies below the minimum $0.440 J_{zz}$ excitation into the exact band complement, so no complement state can occur as a final state in that window. The full Lehmann representation therefore reduces exactly to the following band restriction. The microscopic spin probe is pulled back with the exact zero-source canonical map. Let W_0 denote Eq. (17) at $\boldsymbol{\theta} = 0$ and define

$$\widetilde{S}_a^z(\boldsymbol{q}) = W_0^\dagger \left(\sum_{j \in a} e^{-2\pi i \boldsymbol{q} \cdot \boldsymbol{r}_j} S_j^z \right) W_0, \tag{38}$$

where $a = 0, \ldots, 3$ labels the four pyrochlore sublattices. We use the same $\widetilde{S}_a^z$ with the periodic operator $h(\mathbf{0})$ and the winding-free operator $h_{M=4}$. This avoids attributing a change of probe convention to the winding-free Hamiltonian.

For the low-energy eigenstates of either full Hamiltonian,

$$S^{zz}(\boldsymbol{q}, \omega) = \frac{1}{N d_0} \sum_{g \in \mathcal{G}} \sum_{n \notin \mathcal{G}} \sum_{a=0}^{3} |\langle n | \widetilde{S}_a^z(\boldsymbol{q}) | g \rangle|^2 G_\eta[\omega - (E_n - E_g)], \tag{39}$$

where $\mathcal{G}$ contains the states degenerate with the numerical ground state to $10^{-9}J_{zz}$, $d_0 = |\mathcal{G}|$, and G_η is a normalized Gaussian with $\eta = 6\times10^{-4}J_{zz}$. Equation (39) is the four-sublattice trace of the longitudinal pseudospin response. It is not an unpolarized neutron cross section and includes no local-axis projection, magnetic form factor, or transverse channel.

The four characters of the cubic-16 translation group are represented by

$$\boldsymbol{q} = \Gamma = (0,0,0), \quad X_x = (1,0,0), \quad X_y = (0,1,0), \quad X_z = (0,0,1) \tag{40}$$

in reciprocal cubic units. No interpolated momenta are plotted. The complete line summary is given in Table III. The three winding-free X channels agree to 5×10^{-16} in integrated weight. The Γ weight is suppressed by the projection from 0.3182 to 2.05×10^{-6}. Reconstructing W_0 reproduces the cached periodic operator with relative error 2.84×10^{-14}; its isometry residual is 6.32×10^{-14}. The arrays, unbroadened lines, and diagnostics are stored in `campaign/outputs/dssf_cubic16_p0p046000.npz` and the companion JSON report.

TABLE III. Low-energy cubic-16 longitudinal spectral lines for the periodic and full-Hilbert winding-free Hamiltonians.

Hamiltonian	momentum	d_0	first ω/J_{zz}	integrated weight
periodic ED	Γ	1	0.0806529	0.318193
periodic ED	each X	1	0.0806529	0.212129
winding-free	Γ	6	0.0053438	2.05×10^{-6}
winding-free	each X	6	0.0053335	0.240607

QMC BENCHMARK PROVENANCE AND METRICS

The reference heat capacity and entropy are vector extractions from Fig. 1(b) of Huang *et al.* [1]. The arXiv source contains the original vector PDF, allowing the red C/N and orange S/N paths to be read without raster tracing. The extraction script, axis calibration, CSV, and provenance are stored under `campaign/data/`. Estimated extraction uncertainties are 0.002 in $\log_{10}(T/J_{zz})$, 0.0015 in C/N, and 0.002 in S/N. These are not substitutes for unreported QMC statistical errors.

Curve error is the root-mean-square difference after interpolation in $\log T$ over the digitized support. Thresholds are defined in `campaign/config.json` before evaluation. The current values are listed in Table IV. The converged $M = 4$ winding-free result reduces the low-temperature heat-capacity RMSE by 72.9%, but its final value 0.02327 remains above the preregistered 0.02 tolerance. The entropy gate also fails because $0.04509 > 0.02$. The discrepancy is concentrated in the low-temperature region, where the finite winding-free band retains excess entropy relative to QMC.

TABLE IV. Thermodynamic comparison with the vector-extracted QMC benchmark.

quantity	periodic cubic-16	winding-free cubic-16	QMC
low T_p/J_{zz}	0.02554	0.001463	0.001100
low-T C/N RMSE	0.0859	0.02327	0
high T_p/J_{zz}	0.2549	0.2549	0.2105
S/N RMSE	0.0514	0.04509	0

The same paper specifies a future dynamics gate. It uses an $8\times8\times8$ primitive-cell lattice and reports

$$S^{zz}(\boldsymbol{q},\omega) = \sum_{\alpha=1}^{4} S^{zz}_{\alpha\alpha}(\boldsymbol{q},\omega), \qquad S^{+-}(\boldsymbol{q},\omega) = \sum_{\alpha=1}^{4} S^{+-}_{\alpha\alpha}(\boldsymbol{q},\omega), \tag{41}$$

at $T/J_{zz} = 0.001$, 0.04, and 0.1 along $2\pi(0,2k,0)/8$ and $2\pi(k,k,k)/8$ [1]. ED comparison must use only momenta realized by its finite translation group, the same sublattice trace and spin normalization, a linear frequency axis, and a stated broadening comparable to the QMC-SAC resolution. This matched dynamics calculation has not yet been performed.

COMPLETE VALIDATION STATUS

TABLE V. Current validation gates and the evidence supporting each status.

gate	status	evidence or missing calculation
loop topology	pass	winding 4/6 removed; contractible 6 retained
internal S^{zz} dynamics	pass	periodic and winding-free bands use the same W_0-dressed probe
all-temperature stability	pass	cubic-16 high-T peak position unchanged
QMC heat capacity	fail	exact $M = 4$ low-T RMSE 0.02327 > 0.02
QMC entropy	fail	exact $M = 4$ RMSE 0.04509 > 0.02
character resolution	pass	centered $M = 3 \to 4$ error 0.013%
band overlap	pass	source-resolved minimum is 0.762
order convergence	diagnostic only	production spectrum is exact-band
QMC-SAC dynamics	pending	matched S^{zz} and S^{+-} required
QMC energy	pending	not plotted in the benchmark paper
FCC-32 full Hamiltonian	pending	current result is order-three gauge block
$Ce_2Hf_2O_7$ fit	pending	requires converged FCC-32 ABC/XYZ curves

The remaining production sequence has two deliverables. First, scale the now-general ABC/XYZ source, including its pair-flip channel, to character-converged FCC-32 thermodynamics and dynamics with a stochastic microscopic complement. Second, fit the $Ce_2Hf_2O_7$ heat capacity over those curves. The failed cubic heat and entropy gates and the matched neutron channels become size and holdout tests of the FCC-32 parameter sets; they are not removed from the evidence ledger.

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